

DecodeLabs Data Analytics Internship *(Batch 2026)*

Project 1: Data Cleaning & Preparation Report

Dataset: E-Commerce Orders Dataset

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1. Introduction

This report documents the data cleaning and preparation process carried out as part of Project 1 of the DecodeLabs Data Analytics Internship. The objective of this project was to take a raw, unprocessed e-commerce orders dataset and transform it into a clean, reliable, and analysis-ready dataset by identifying and resolving common data quality issues.

Following the principles outlined in the project guidelines, the focus was on data integrity rather than visualization: identifying missing values, eliminating duplicate records, and correcting inconsistent formatting in dates and numeric fields.

2. Dataset Overview

The dataset used for this project is an e-commerce orders dataset containing transactional records with information about products, customers, pricing, payment methods, and order status.

Attribute	Value
Dataset Name	E-Commerce Orders Dataset
Total Records (Rows)	1,200
Total Fields (Columns)	14
Date Range Covered	January 2023 – June 2025
Key Fields	OrderID, Date, Product, Quantity, UnitPrice, TotalPrice, OrderStatus

3. Methodology

The cleaning process followed a structured, three-phase approach as recommended in the project guidelines:

3.1 Phase 1: Strategic Imputation (Handling Missing Values)

Rather than deleting rows with missing values — which would reduce the statistical power of the dataset — missing entries in the CouponCode column were identified and filled with a meaningful placeholder label, “NO_COUPON”. This preserves the integrity of the record while clearly indicating that no coupon was applied to that order.

3.2 Phase 2: Integrity Audit (Duplicate Check)

Each record was checked for exact duplicate rows and for duplicate OrderID values, since each order should be a unique transaction. The audit confirmed that the dataset contained no duplicate records and no duplicate OrderIDs, meeting the “zero duplicate IDs” requirement for the project.

3.3 Phase 3: Format Standardization (Speak One Language)

To ensure consistency across the dataset, the following formatting corrections were applied:

- The Date column was standardized to the ISO 8601 format (YYYY-MM-DD) for consistency and compatibility with analytical tools.
- Numeric values in the TotalPrice column that contained floating-point precision errors (e.g., values with more than two decimal places) were rounded to two decimal places using the ROUND() function.

4. Tools Used

Microsoft Excel was used to perform all data cleaning operations, including Find & Replace for handling missing values, Format Cells for date standardization, and the ROUND() formula for numeric precision correction. This approach is suitable for the dataset size of 1,200 records, in line with the project guidance that Excel is appropriate for datasets under one million rows.

5. Issues Identified and Resolved

Issue Identified	Affected Records	Resolution Applied
Missing values in CouponCode column	309 of 1,200 rows	Imputed with “NO_COUPON” label
Inconsistent date representation in Date column	All 1,200 rows	Standardized to YYYY-MM-DD format

Floating-point precision errors in TotalPrice	Approx. 50 rows	Rounded to 2 decimal places using ROUND()
Duplicate records / duplicate OrderIDs	0 found	Verified — no action required

6. Change Log

In line with the project requirement that “if it isn’t documented, it didn’t happen,” the following change log records every modification made to the dataset, along with its impact and resolution status.

Change ID	Description	Impact	Status
CR001	Filled missing CouponCode values with “NO_COUPON” label	309 records updated	Resolved
CR002	Standardized Date column to ISO 8601 format (YYYY-MM-DD)	1200 records standardized	Resolved
CR003	Rounded TotalPrice values to 2 decimal places for numeric precision	Approx. 50 records corrected	Resolved
CR004	Verified zero duplicate OrderIDs across the dataset	1200 records verified	Resolved

7. Conclusion

Through this project, the raw e-commerce orders dataset was successfully transformed into a clean, consistent, and analysis-ready dataset. Missing values were handled through strategic imputation rather than deletion, duplicate records were audited and confirmed absent, and inconsistencies in date and numeric formatting were corrected to meet the verification gate requirements (0% error rate on unique identifiers and date formats).

This project provided practical, hands-on experience in the foundational skills of data analytics — data cleaning and preparation — which form the basis for all further analysis and form a key part of a professional data analyst's portfolio.